

JEE MAIN REVISION MASTER: IONIC EQUILIBRIUM

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 8 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	pH Calculation of Acids, Bases & Salts	5 PYQs	pH & Hydrolysis
Moderate	Solubility Product (Ksp) & Common Ion Effect	3 PYQs	Solubility

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
pH Calculation of Acids, Bases & Salts	$\text{pH} = \text{p}K_a + \log \frac{[\text{conjugate base}]}{[\text{acid}]}, \quad \text{Salt of WA-SB: } \text{pH} = 7 + \frac{1}{2} \text{p}K_a + \frac{1}{2} \log c$
Solubility Product (Ksp) & Common Ion Effect	$A_x B_y \rightleftharpoons x A^{y+} + y B^{x-} \implies K_{sp} = x^x y^y s^{x+y}, \quad Q_{sp} > K_{sp} \implies \text{Precipitation occurs}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 pH Calculation of Acids, Bases & Salts (Moderate Yield — 5 PYQs)

Core Principle: Self-ionization of water ($K_w = 10^{-14}$ at 25°C), Ostwald dilution law, buffer solutions, and salt hydrolysis formulas.

■ **Caution / JEE Pitfall:** pH of 10^{-8} M HCl is NOT 8 (cannot be basic!); including water's auto-ionization $[H^+] = 10^{-8} + 10^{-7}$ implies $\text{pH} \approx 6.79$.

■ **Speed Trick:** Buffer capacity is maximum when $[\text{salt}] = [\text{acid}] \implies \text{pH} = \text{p}K_a$.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "A 0.15 mole of pyridinium chloride has been added to 500cm³ of 0.2 M pyridine solution (a base). Assuming there is no change..." **[Ans: N/A]**
- **[2023 Shift] Q#3 (MCQ):** "A weak acid with $\text{p}K_a$ 5.9 and weak base with $\text{p}K_b$ 5.8 are mixed in equal proportions. pH of the resulting solution is..." **[Ans: N/A]**
- **[2022 Shift] Q#4 (MCQ):** "Which among the following has highest pH ?..." **[Ans: N/A]**

3.2 Solubility Product (K_{sp}) & Common Ion Effect (Moderate Yield — 3 PYQs)

Core Principle: Calculation of molar solubility s from K_{sp} , ionic product comparison, and suppression of solubility via common ion effect.

■ **Caution / JEE Pitfall:** For Ag_2CrO_4 (A₂B type): $K_{sp} = (2s)^2(s) = 4s^3$ implies $s = \sqrt[3]{(K_{sp})/4}$, NOT s^2 !

■ **Speed Trick:** Common ion effect ALWAYS suppresses the solubility of a sparingly soluble salt (e.g. AgCl is less soluble in NaCl than pure water).

Representative Source PYQs:

- [2024 Shift] Q#2 (MCQ): "Solubility product of CaC_2O_4 at a given temperature in pure water is $4 \times 10^{-9}(\text{molL}^{-1})^2$. Solubility of CaC_2O_4 ..." [Ans: N/A]
- [2020 Shift] Q#2 (MCQ): "The conjugate base of NH_3 is..." [Ans: N/A]
- [2019 Shift] Q#4 (MCQ): "Solubility of AgCl is least in..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A 0.15 mole of pyridinium chloride has been added to 500cm³ of 0.2 M pyridine solution (a base). Assuming there is no change in volume upon mixing (Given: pK_b of pyridine = 8.82), the pH of the resulting solution is

- (A) 5
- (B) 6
- (C) 7
- (D) 8

Correct Answer: None

Solution Analysis: Moles of pyridine present initially: $0.2 \times (500 / 1000) = 0.1 \text{ mol}$ Moles of pyridinium chloride added: 0.15 mol Pyridine is a weak base, and pyridinium chloride gives its conjugate acid, pyridinium ion. So the mixture acts as a basic buffer. For a basic buffer, $\text{pOH} = \text{pK}_b + \log([\text{salt}] / [\text{base}])$ Since volume is the same for both, we can directly use moles: $\text{pOH} = 8.82 + \log(0.15 / 0.10)$ $\text{pOH} = 8.82 + \log(1.5)$ $\log(1.5) \approx 0.18$ So, $\text{pOH} = 8.82 + 0.18 = 9.00$ Now, $\text{pH} = 14 - \text{pOH}$ $\text{pH} = 14 - 9 = 5$ So the pH of the result

Benchmark Problem 2: KCET 2024 (Q#2)

Solubility product of CaC_2O_4 at a given temperature in pure water is $4 \times 10^{-9}(\text{molL}^{-1})^2$. Solubility of CaC_2O_4 at the same temperature is

- (A) $6.3 \times 10^{-5} \text{ molL}^{-1}$
- (B) $2 \times 10^{-5} \text{ molL}^{-1}$
- (C) $2 \times 10^{-4} \text{ molL}^{-1}$
- (D) $6.3 \times 10^{-4} \text{ molL}^{-1}$

Correct Answer: None

Solution Analysis: Given, $K_{sp} = 4 \times 10^{-9}(\text{molL}^{-1})^2$ $S = ?$

$\text{CaC}_2\text{O}_4 \rightleftharpoons \text{Ca}^{2+} + \text{C}_2\text{O}_4^{2-}$ $4 \times 10^{-9} = S^2 \Rightarrow S = \sqrt{4 \times 10^{-9}} = 6.3 \times 10^{-5} \text{ mol/L}$
Thus, the solubility of CaC_2O_4 is $6.3 \times 10^{-5} \text{ mol/L}$.